

## Motivation and research question

My research motivation stems from the observation that, without an explicit “minimal correction” prompt, LLMs tend to completely rewrite sentences written by non-native speakers to conform to the expression style of native speakers (Kopylova et al., 2026; Munda et al., 2024). In contrast, BabyLM (Warstadt et al., 2023) has lower cognitive complexity, and I hypothesize that it may align more naturally with the “minimal correction” teaching principle, thereby avoiding “overcorrection.” However, the experimental results involving BabyLMs indicate that the non-instruction-tuned models are unable to perform this grammatical error correction task. Consequently, my research question shifted to: Does the number of parameters in locally deployable, open-source, and free large language models affect the phenomenon of overcorrection when the models correct grammatical errors?

## Experimental setup

The corpus I used was: W&I+LOCNESS v2.1 (BEA-2019; Bryant et al., 2019), CEFR Level A dev split, A.dev.gold.bea19.m2. From the corpus’s 1,037 sentences, I kept those that range from 5 to 40 tokens and contain at least one non-noop edit, excluding punctuation- and orthography-only changes and 14 degenerate items (where the recorded correction was identical to the source). This left 705 sentences, from which I randomly sampled 200 (seed 42); first 20 served as the pilot sample. Gold corrections were reconstructed by applying M2 edit operations (see `prepare_data.py`); manually verified, 0 reconstruction errors.

I ran the test program on the pilot sample and inspected the results. To determine whether a model made the correct corrections, I used ERRANT (Bryant et al., 2017) and computed recall from its output. However, I discovered during the experiment that, since both GPT-2 (Radford et al., 2019) and GPT-BERT (Charpentier & Samuel, 2024) are base (non-instruction-tuned) models, they tended to copy or continue the input in the 20-sentence sample test. Consequently, I had to change the test subjects.

So I selected large language models from the same family Qwen2.5 (Qwen Team, 2025), with only the number of parameters as the independent variable, to minimize differences between the models under study. Using the same experimental procedure, the 20-sentence sample test ran smoothly this time, with no instances of copying or continuing the input. Since all the Qwen2.5 models are instruction-tuned, I used the following zero-shot prompt: “Correct only grammatical errors in this sentence and then only output the corrected sentence.” This resulted in 200 items  $\times$  3 models = 600 generations, with 0 failures. I manually removed the preface such as “The corrected sentence is:” from the outputs (0.5B 7/200, 1.5B 15/200, 3B 0/200) and suffix “(No grammatical errors to correct)” (0.5B 0/200, 1.5B 2/200, 3B 5/200).

## Hypotheses

I treat overcorrection as the normalized edit distance plus the number of FP. My hypothesis is that large language models with fewer parameters are less prone to overcorrection.

## Results

Finding 1: As the number of parameters increases, the model becomes less prone to overcorrection; however, this decrease follows a “stepwise” rather than a “gradual” pattern.

| Condition       | median | 95% CI         |
|-----------------|--------|----------------|
| Human annotator | 0.167  | [0.143, 0.200] |
| Qwen2.5-0.5B    | 0.244  | [0.210, 0.250] |

|              |       |                |
|--------------|-------|----------------|
| Qwen2.5-1.5B | 0.231 | [0.194, 0.263] |
| Qwen2.5-3B   | 0.167 | [0.143, 0.194] |

The median values in this table tell us how much each condition modifies a typical sentence (normalized edit distance). It indicates that 3B makes fewer changes than the two smaller models.

Friedman  $\chi^2(2) = 25.64$ ,  $p < .001$ . The three models therefore differ significantly

Using the Wilcoxon signed-rank test (Holm-corrected) shows the differences between pairs of models:

0.5B vs 1.5B  $p_{\text{holm}} = .57$ ,  $r = .10$ . 0.5B vs 3B  $p < .001$ ,  $r = .53$ . 1.5B vs 3B  $p < .001$ ,  $r = .56$ .

0.5B and 1.5B both differ from human ( $p < .001$ ,  $r = .50 / .46$ ); 3B vs human  $p_{\text{holm}} = 0.57$ ,  $r = .09$ . Both small models make many more changes than human annotator.

| Model        | TP  | FP  | FN  | Prec.  | Rec.   | F0.5   | F0.5 without ORTH |
|--------------|-----|-----|-----|--------|--------|--------|-------------------|
| Qwen2.5-0.5B | 121 | 887 | 432 | 0.1200 | 0.2188 | 0.1320 | 0.1613            |
| Qwen2.5-1.5B | 163 | 900 | 390 | 0.1533 | 0.2948 | 0.1696 | 0.2078            |
| Qwen2.5-3B   | 181 | 724 | 372 | 0.2000 | 0.3273 | 0.2169 | 0.2887            |

The edit magnitude is nearly the same for 0.5B and 1.5B, but it drops sharply at 3B. Based on these statistics, I conclude that there is a threshold between 1.5B and 3B, rather than a continuous pattern where “the more parameters, the fewer corrections.”

Finding 2: While fixing more actual errors, 3B made fewer unnecessary changes.

Compared to 0.5B, 3B shows an increase in TP and a decrease in FP. This indicates that 60 more correct changes were made and 163 fewer unnecessary changes were made. Most of its changes are still unnecessary (precision 0.20).

Finding 3: ORTH is a constant artefact and should be excluded

FP values from ORTH: 251 / 251 / 278 — these are essentially consistent across different conditions and therefore do not reflect the model’s behavior. These are differences in capitalization and word forms (what -> What, It's -> It's), i.e., issues with the output format rather than error correction behavior.

Based on the experimental results, I can reject my hypothesis; in fact, large language models with more parameters are less prone to overcorrection.

Repository: <https://github.com/whks-wu/llm-seminar-project.git>

## References

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